GenePair: Statistical Methods for Modeling Spatially-Referenced Paired Genetic Relatedness Data

Transmission Probabilities Model

$$Y_{ij} \overset{\text{ind}}{\sim} f_{y_{ij}}(y) = (1-\pi_{ij})^{1(y=0)} \left[\frac{\pi_{ij}}{y(1-y)} f_{w_{ij}} \left\{ \ln\left(\frac{y}{1-y} \right) \right\} \right]^{1(y>0)}, \; y \in [0,1), \; i = 1,...,n, \; j = 1,...,n, \; i \neq j$$

- $$\ln\left(\frac{\pi_{ij}}{1-\pi_{ij}} \right) = \mathbf{x}_{ij}^{\mathrm{T}} \boldsymbol{\beta}_z + \mathbf{d}_j^{\mathrm{T}} \boldsymbol{\gamma}_z^{(g)} + \mathbf{d}_i^{\mathrm{T}} \boldsymbol{\gamma}_z^{(r)} + \theta_{zj}^{(g)} + \theta_{zi}^{(r)} + \nu_{zi}\nu_{zj}$$
- $$f_{w_{ij}}(w) \equiv \mathrm{N}\left(\mathbf{x}_{ij}^{\mathrm{T}} \boldsymbol{\beta}_w + \mathbf{d}_j^{\mathrm{T}} \boldsymbol{\gamma}_w^{(g)} + \mathbf{d}_i^{\mathrm{T}} \boldsymbol{\gamma}_w^{(r)} + \theta_{wj}^{(g)} + \theta_{wi}^{(r)} + \nu_{wi}\nu_{wj}, \; \sigma_\epsilon^2 \right)$$

$$\begin{aligned}
\theta_{zi}^{(g)} &= \eta_z^{(g)} \{ d(\mathbf{s}_i) \} + \zeta_{zi}^{(g)}, \\
\theta_{zi}^{(r)} &= \eta_z^{(r)} \{ d(\mathbf{s}_i) \} + \zeta_{zi}^{(r)}, \\
\theta_{wi}^{(g)} &= \eta_w^{(g)} \{ d(\mathbf{s}_i) \} + \zeta_{wi}^{(g)}, \\
\theta_{wi}^{(r)} &= \eta_w^{(r)} \{ d(\mathbf{s}_i) \} + \zeta_{wi}^{(r)}, \; i = 1,...,n,
\end{aligned}$$

$$\begin{aligned}
&\boldsymbol{\eta} | \Omega, \phi \sim \mathrm{MVN}\{ \mathbf{0}_{4m}, \Sigma(\phi) \otimes \Omega \} \text{ where} \\
&\Sigma(\phi)_{ij} = \exp\left\{ -\phi \left\| \mathbf{s}_i^* - \mathbf{s}_j^* \right\| \right\}, \\
&\boldsymbol{\eta}^{\mathrm{T}} = \left\{ \boldsymbol{\eta}(\mathbf{s}_1^*)^{\mathrm{T}}, \ldots, \boldsymbol{\eta}(\mathbf{s}_m^*)^{\mathrm{T}} \right\}, \text{ and} \\
&\boldsymbol{\eta}(\mathbf{s}_i^*)^{\mathrm{T}} = \left\{ \eta_z^{(g)}(\mathbf{s}_i^*), \eta_z^{(r)}(\mathbf{s}_i^*), \eta_w^{(g)}(\mathbf{s}_i^*), \eta_w^{(r)}(\mathbf{s}_i^*) \right\}.
\end{aligned}$$

- $d(\mathbf{s}_i)$: Maps the spatial location of an individual to an entry within a smaller set of $m < n$ unique locations such that $d(\mathbf{s}_i) \in \{\mathbf{s}_1^*, \ldots, \mathbf{s}_m^*\}$;
- $\zeta_{zi}^{(g)} | \sigma^2_{\zeta_z^{(g)}} \overset{\text{iid}}{\sim} \mathrm{N}\left(0, \sigma^2_{\zeta_z^{(g)}}\right)$, $\zeta_{zi}^{(r)} | \sigma^2_{\zeta_z^{(r)}} \overset{\text{iid}}{\sim} \mathrm{N}\left(0, \sigma^2_{\zeta_z^{(r)}}\right)$, $\zeta_{wi}^{(g)} | \sigma^2_{\zeta_w^{(g)}} \overset{\text{iid}}{\sim} \mathrm{N}\left(0, \sigma^2_{\zeta_w^{(g)}}\right)$, $\zeta_{wi}^{(r)} | \sigma^2_{\zeta_w^{(r)}} \overset{\text{iid}}{\sim} \mathrm{N}\left(0, \sigma^2_{\zeta_w^{(r)}}\right)$;
- $\nu_{zi} | \sigma^2_{\nu_z} \overset{\text{iid}}{\sim} \mathrm{N}\left(0, \sigma^2_{\nu_z}\right)$, $\nu_{wi} | \sigma^2_{\nu_w} \overset{\text{iid}}{\sim} \mathrm{N}\left(0, \sigma^2_{\nu_w}\right)$, $i = 1,...,n$;
- m: Number of unique spatial locations ($m \leq n$);
- n: Number of individuals.

Prior Information

$\beta_{zj}, \beta_{wj}, \gamma_{zk}^{(g)}, \gamma_{zk}^{(r)}, \gamma_{wk}^{(g)}, \gamma_{wk}^{(r)} \overset{\text{iid}}{\sim} \mathrm{N}\left(0, \sigma_r^2\right), \; j = 1,...,p_x, \; k = 1,...,p_d$;

- p_x: Length of $\mathbf{x}_i$ vector (same for all i) which includes an intercept term;
- p_d: Length of $\mathbf{d}_j$ vector (same for all j) which **does not** include an intercept term;
- Default setting: $\sigma_r^2 = 10,000$.

$\sigma_\epsilon^2 \sim \text{Inverse Gamma}\left(a_{\sigma_\epsilon^2}, b_{\sigma_\epsilon^2}\right)$;

- Default setting: $a_{\sigma_\epsilon^2} = 0.01$, $b_{\sigma_\epsilon^2} = 0.01$.

$\sigma^2_{\zeta_z^{(g)}} \sim \text{Inverse Gamma}\left(a_{\sigma^2_{\zeta_z^{(g)}}}, b_{\sigma^2_{\zeta_z^{(g)}}}\right)$, $\sigma^2_{\zeta_z^{(r)}} \sim \text{Inverse Gamma}\left(a_{\sigma^2_{\zeta_z^{(r)}}}, b_{\sigma^2_{\zeta_z^{(r)}}}\right)$, $\sigma^2_{\zeta_w^{(g)}} \sim \text{Inverse Gamma}\left(a_{\sigma^2_{\zeta_w^{(g)}}}, b_{\sigma^2_{\zeta_w^{(g)}}}\right)$,
$\sigma^2_{\zeta_w^{(r)}} \sim \text{Inverse Gamma}\left(a_{\sigma^2_{\zeta_w^{(r)}}}, b_{\sigma^2_{\zeta_w^{(r)}}}\right)$;

- Default setting: $a_{\sigma^2_{\zeta_z^{(g)}}}, a_{\sigma^2_{\zeta_z^{(r)}}}, a_{\sigma^2_{\zeta_w^{(g)}}}, a_{\sigma^2_{\zeta_w^{(r)}}} = 0.01$, $b_{\sigma^2_{\zeta_z^{(g)}}}, b_{\sigma^2_{\zeta_z^{(r)}}}, b_{\sigma^2_{\zeta_w^{(g)}}}, b_{\sigma^2_{\zeta_w^{(r)}}} = 0.01$.

$\sigma^2_{\nu_z} \sim \text{Inverse Gamma}\left(a_{\sigma^2_{\nu_z}}, b_{\sigma^2_{\nu_z}}\right)$; $\sigma^2_{\nu_w} \sim \text{Inverse Gamma}\left(a_{\sigma^2_{\nu_w}}, b_{\sigma^2_{\nu_w}}\right)$;

- Default setting: $a_{\sigma^2_{\nu_z}}, a_{\sigma^2_{\nu_w}} = 0.01$, $b_{\sigma^2_{\nu_z}}, b_{\sigma^2_{\nu_w}} = 0.01$.

$\Omega^{-1} \sim \text{Wishart}\left(\nu, \Omega^*\right)$;

- Default setting: $\nu = 5$, $\Omega^* = I_4$.

$\phi \sim \text{Gamma}\left(a_\phi, b_\phi\right)$;

- Default setting: $a_\phi = 1.00$, $b_\phi = 1.00$.

Default Initial Values

- $\beta_{zj}, \beta_{wj}, \gamma^{(g)}_{zk}, \gamma^{(r)}_{zk}, \gamma^{(g)}_{wk}, \gamma^{(r)}_{wk} = 0$ for all j, k;
- $\theta^{(g)}_{zi}, \theta^{(r)}_{zi}, \theta^{(g)}_{wi}, \theta^{(r)}_{wi} = 0$ for all i;
- $\eta^{(g)}_{zi}, \eta^{(r)}_{zi}, \eta^{(g)}_{wi}, \eta^{(r)}_{wi} = 0$ for all i;
- $\sigma^2_\epsilon = \text{variance}\left(\boldsymbol{Y}\right)$;
- $\sigma^2_{\zeta_z^{(g)}}, \sigma^2_{\zeta_z^{(r)}}, \sigma^2_{\zeta_w^{(g)}}, \sigma^2_{\zeta_w^{(r)}} = 1.00$;
- $\sigma^2_{\nu_z}, \sigma^2_{\nu_w} = 1.00$;
- $\Omega = I_4$;
- $\phi = 1.00$.

Additional Information

- v: Required input; an n by m matrix with $\text{v}_{ij} = 1$ if individual i resides at location $\mathbf{s}^*_j$ and $\text{v}_{ij} = 0$ otherwise;
- It is recommended to scale the matrix of spatial distances by the largest observed distance in the matrix prior to running the model.